1. The blue color of the sky is primarily derived from what type of scattering? What conditions are required for this type of scattering to occur? (2)

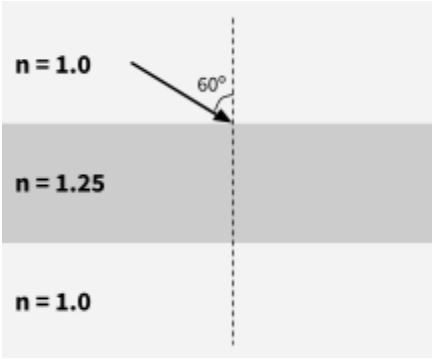
   Rayleigh scattering (1), when the particle size is much smaller than the wavelength of the light (1)

2. Identify the most appropriate method of imaging for each of the following scenarios, choosing from: I) Active sensing II) Whisk broom passive sensing III) Push broom passive sensing; each term may be used multiple times or none at all. (4)

   III An aircraft is rapidly flying over a field and wants to image the vegetation underneath.
   I A satellite in geostationary orbit wants to vertically profile clouds in order to predict rainfall.
   II Bob wants to build his own remote sensor. He is financially limited but wants a flexible sensor that can scan various specialized geometry or terrains.
   I Alice also wants a remote sensor, but she wants whichever option that can provide the highest resolution.

3. What are two ways that remote sensing can be used in forecasting volcanic eruptions? (2)
   Accept any two of the following, 2 points max:
   - Measuring the deformation of the ground (0.5) through altimetry or radar or other methods (0.5)
   - Measuring changes in surface temperature (0.5) with radiometers or pyrometers/thermometers (0.5)
   - Measuring gas emission/content (0.5) through spectrometers (0.5)

4. A beam of light enters from air (index of refraction = 1.0) to a layer of peculiar air with index of refraction of 1.25 at an angle of 60º from the normal. At what angle from the normal would the light beam emerge from the peculiar air back into air, if at all? Show your work or seasoning. (3)



   The angle is still 60º (1), because the initial and final media have the same indices of refraction OR shows work for arriving at 60º (2)

Use Image 1, which depicts a region of Taiwan, for the following section.

5. What units are the margins of the map labeled with? (1)

Degrees (0.5), arcminutes or minutes of arc or minutes (0.5)

6. What remote sensing technique is primarily used to obtain this image? Briefly explain how this technique is able to measure the elevation of the ground. (3)

Radar altimetry (1), it sends pulses towards the ground (1) and measures the time it takes the pulse to bounce off the ground back to the sensor (1).

7. What region of the electromagnetic spectrum does this technique primarily operate in? (1)

Microwave

8. Name something else that this technique could be good for measuring. (1)

Cloud or precipitation profiles, anti-aircraft guns, probing beneath the ground/ice, anything else radar/radar altimeters can be reasonably used for

9. What is the overall trend of this region's change in elevation? (1)

a. Increase
b. Decrease
c. Constant

10. What is a likely anthropogenic cause for the change you selected in the previous question? Briefly explain the mechanism through which it occurs. (3)

Groundwater extraction faster than the rate of replenishment (1), causing aquifer pores to empty and collapse (water is more incompressible than air), lowering surface elevation (2)

11. List two natural and two anthropogenic sources of aerosols on Earth. (2)

0.5 for each listed, max 1 point in each category:
Natural: volcanic eruptions, forest fires, wind blowing dust into the air, plant/fungal emissions, sea spray, anything else reasonable
Anthropogenic: burning fossil fuels (including transportation, energy generation, etc.), refrigerants, cloud seeding

12. Refer to Image 2, which depicts aerosol hotspots across the globe during January and July. Match the months to each image and give one reason of how you can tell. (2)

2a: July
2b: January (1 pt for both correct)
Accept any of the following: monsoon season in northern hemisphere leading to more aerosols, ITCZ is further north in 2a (indicating northern hemisphere warm season), more human activity or plant emissions during summer, more fires tend to occur during summer, or the data range covers more of northern hemisphere in 2a indicating the sun is more directly over the northern hemisphere (and the satellite is sun synchronous).

13. What instrument on Terra took these images? How does it physically measure aerosol concentration? (2)

MODIS (Moderate Resolution Imaging Spectroradiometer) (1), it measures aerosol optical depth, which is computed from the spectral analysis reflected sunlight (1)

14. Explain the Twomey effect. Be sure to explain how the addition of aerosols influences clouds and ultimately Earth's radiative balance. (3)

The addition of aerosols to the atmosphere lead to smaller droplet sizes (by acting as condensation nuclei) (1), causing clouds to reflect more sunlight (1) and cools the Earth (1).

15. Explain how the **opposite** of what you described above might be true (e.g. if you previously concluded the Twomey effect states aerosols **cool** the Earth, now propose a process by which aerosols can **warm** the Earth, and vice versa). (2)

Dark aerosols such as dust or soot (1) absorb heat and reradiate it to heat up the surrounding atmosphere (1).

Images 3a, 3b and 3c display a Bay Area region in multiple filters, with data from Sentinel-2. Use Images 3a-c to answer the following section (please hold off on referring to 3d and 3e until further instructed).

16. How is false color implemented in the context of remote sensing (as opposed to true color)? Which image(s) between 3a-c display(s) false coloration, if any? (3)

Colors assigned to the image are not what they directly appear as in real life, especially for non-optical radiation such as infrared (2). 3b and 3c depict false coloration (1 for both).

17. What orbit is Sentinel-2 in and why might that be helpful for its task? (2)

Sun-synchronous (1), passes over the same region at the same local solar time to ensure consistent lighting conditions for comparison (1).

18. Sentinel-2 has a spatial resolution of 10 meters in Image 3a. A square of what area is covered by each pixel? Assuming the images provided are 32.5 km by 53 km, what is the overall resolution of the Image 3a in pixels? Show your work. (4)

$(10m)^2 = 100\ m^2$ is covered by each pixel (2), $32500m*53000m/100m^2 = 17225000$ pixels OR $32500/10*53000/10 = 3250$ pixels x 5300 pixels (1 pt for work, 1 pt for solution)

19. Match the following descriptions of Sentinel-2 to the type of resolution being described, choosing from: Temporal, Radiometric, Spectral. (3)

Radiometric	Sentinel-2's 12-bit resolution allows it to distinguish between 4,096 levels of brightness
Spectral	Sentinel-2 has 13 bands, covering the electromagnetic spectrum from 442 nm to 2202.4 nm
Temporal	Sentinel-2 revisits the same location every 5 days

20. What general region of the electromagnetic spectrum is used to produce Image 3a? (1)
Visible

21. Are the clouds displayed near the top right of Image 3a high or low clouds? Do you expect them to contribute to a net warming or cooling of the surface? Explain. (3)

High clouds (1), net warming (1) because they are thin and reflect light poorly (0.5) but absorb and reradiate much of the outgoing longwave radiation (0.5)

22. In Image 3a , what color generally corresponds to the densest vegetation? (1)

a. Brown
b. Yellow
c. White
d. Blue
e. Green

23. In Image 3b, what color generally corresponds to the densest vegetation? (1)

a. Red
b. Yellow
c. Orange
d. Light green
e. Dark green

24. In Image 3c, what generally color corresponds to the densest vegetation? (1)

a. White
b. Blue
c. Light green
d. Dark green

25. Which of the labeled points on the surface has the highest albedo? (1)

a. A
b. B
c. C
d. D

26. Place the following electromagnetic regions on the spectrum provided: near infrared (NIR), far infrared (FIR), short-wave infrared (SWIR), mid-wave infrared (MWIR), long-wave infrared (LWIR). (2)

Visible _____ _____ _____ _____ _____ Microwave

NIR, SWIR, MWIR, LWIR, FIR (-0.5 for each wrong placement)

27. Image 3b utilizes the NDVI to display vegetation health. What does NDVI stand for? (1)
Normalized Difference Vegetation Index

28. The NDVI calculation requires the visible red and what region of the infrared spectrum as listed in the previous question? What bands do they correspond to, respectively? (2)

NIR (1); red = B04 (0.5), NIR = B08 (0.5)

29. The data utilized in NDVI primarily relies on which of the following interactions of light with vegetation? (1)

a. Reflection
b. Refraction
c. Diffraction
d. Scattering

30. Image 3c utilizes the NDWI, a similar index for water instead of vegetation. It can be calculated using the following formula for Sentinel-2: NDWI = (Band 3 - Band 8) / (Band 3 + Band 8). Circle the words that best complete the following sentence: (2)

If Band 3 receives a much stronger signal than Band 8, the NDWI will be **(low/high)**, indicating **(water/dry land)**. Band 8 is **(longer/shorter)** in wavelength than Band 3, as water produces stronger signal in **(longer/shorter)** wavelengths compared to dry land.

31. What is the range of all possible values for the NDWI? In other words, what are the lowest and highest possible NDWI values? (1)
-1 to 1

32. What color in image 3c has the lowest NDWI value? (1)
White

33. Which of the following best identifies the features circled in E on the maps? (1)

a. Greenbelts
b. Buildings
c. Crops field
d. Artificial ponds

34. Use evidence from at least 2 images between 3a-c to support your choice above. (2)

1 point for any valid explanation given, 2 points max; if d was not selected for #33, accept answers as appropriate to the selected answer:

3a: dark color/low albedo can indicate water body

3b: same color (red) as the bay/water bodies; poor in vegetation or low NDVI

3c: Mostly blue in color, which is the same as water bodies; high NDWI

35. Other than point B, which labeled point/region would you expect to have the highest risk of flooding (choose from A, C, D, E)? Briefly explain your answer. (2)

Point C (1), located at a low elevation (likely a wetland) and is very close to water bodies (1)

36. At which labeled point would you expect to have the lowest variation in daily temperature? Why? (2)

Point B (1), as water has high heat capacity and acts as a buffer for temperature changes (1).

37. Notice the main river on the left. Which general direction is it flowing in? (1)

a. Northeast
b. Northwest
c. Southeast
d. Southwest

38. Notice that the majority of the branches of the river are to the west side of the river. Why might that be the case? (2)

The elevation increases relatively steeply (there is a hill) to the east (1) and prevents meandering channels from developing, unlike the west that is relatively flat (1).

39. Refer to image Image 4d, which shows the annual NLCD land classification by the USGS in 2024. Assign a Land Cover Class, as listed, to each labeled point in Image 4d. (4)

a. Developed
b. Barren
c. Herbaceous
d. Shrubland
e. Cultivated
f. Wetland

f Point W
c Point X
e Point Y
a Point Z

40. Compare Image 4d with 4e, which is the NLCD classification for 2004. What are two differences that you notice? (2)

1 point for any valid observation, including ones that are relevantly noticeable but not listed: Part of the wetland has become cultivated; urbanization in the northeast corner; drying of some permanently flooded areas to become wetlands; areas of water body becoming barren (likely manmade constructions)

Tiebreaker: What is the CO2 level today (as seen on co2.earth)? Provide your answer in ppm to at least two decimal places. (0)
Refer to co2.earth if tiebreaking is necessary.